MAN HEI CHENG

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RESEARCH FOCUS

Research in near-term and early fault-tolerant quantum computing, with emphasis on noise-aware quantum algorithms, Clifford and stabilizer structure, mid-circuit measurements, and resource-efficient simulation of fermionic systems. Experienced in numerical benchmarking, Pauli noise propagation, and tensor-network methods with relevance to near-term applications

EDUCATION AND HONOURS

Imperial College London

Ph.D. in Physics *Oct 2021–May 2026*

Supervisor: Prof. M. S. Kim

Thesis Title: Initialization, Encoding and Error Characterization for Near-Term Quantum Computation funded by Fraunhofer Institute ITWM (industry-sponsored doctoral studentship)

MSc. with Distinction in Quantum Fields and Fundamental Forces *Oct 2020–Oct 2021*

BSc. with First Class Honours in Physics with Theoretical Physics *Oct 2017–Oct 2020*

Dean's List 2018

TECHNICAL SKILLS

Programming Languages: Python, Julia, MATLAB (scientific computing, numerical simulation)

Quantum Software: Fermionic simulation, variational algorithms, QSCI methods, Clifford circuits and stabilizers, Pauli propagation in the scope of noise characterization

Quantum Hardware: IBM hardware benchmarking, quantum–classical interface

Classical Computing: Linux, HPC workflows

RESEARCH AND TEACHING EXPERIENCE

Internship in Algorithmiq Oy *Aug–Nov 2024*

Junior Researcher, manager: Dr. S. Filippov *Helsinki, Finland*

- Developed a new generalized cycle benchmarking framework to assess bottlenecks in quantum gate.
- Improved and generalized existing noise-characterization techniques using tensor-network channel tomography.

Visiting Researcher, City University of Hong Kong *Jul–Sep 2023*

Visitor, supervisor: Prof. Min-Hsiu Hsieh and Alice Hu *Hong Kong SAR*

- Initiated a research program on optimal qubit-compression schemes with different conserved quantities.
- Oversaw the experimental realization of an optimal fermionic-encoding scheme.

Visiting Student, Fraunhofer Institute ITWM *Jan–Jun 2023*

Exchange student, supervisor: Dr. V. Bartsch and Dr. A. C. Medina *Kaiserslautern, Germany*

- Benchmarked quantum resource requirements for molecular simulation.
- Wrote a research proposal for the Competence Center Quantum Computing Rhineland-Palatinate.

Visiting Researcher, Korean Institute of Advanced Studies *Aug–Sep 2022*

Visitor, supervisor: Prof. Hyukjoon Kwon *Seoul, Korea*

- Engaged with experts in material design and quantum chemistry to identify key challenges in material simulation.

Co-Mentorship of Ph.D. Research Students *Aug 2025–Present*

Mentor, student: Duheon Song *London, United Kingdom*

- Conceived a new coherent-noise channel magnification framework beyond standard quantum tomography.
- Co-mentoring a first-year PhD student on a new research direction in generalized coherent noise characterization, including theoretical modeling, literature review, and computational foundations.

Teaching Assistant, Quantum Information (Imperial College London) — *Academic Year 22/23*
General teaching assistant — *London, United Kingdom*

- Assisted with graduate-level instruction in quantum information theory and taught technical approaches to solving problems in quantum information.
- Organized feedback sessions to address students' questions during exam preparation.

MSc. Project Supervision: Feature Space Analysis of VQE — *Oct 2022–Oct 2023*
Mentor, students: Aaron Thomas and Vasanth Balla — *London, United Kingdom*

- Supervised master's students in analyzing ansatz design via the feature space induced by an ansatz-defined quantum kernel, using machine-learning methods.
- Mentored students on literature review, cost-function formulation, and technical implementation.

PROJECTS AND PUBLICATIONS

Clifford Circuit Initialization for Variational Quantum Algorithms — *[Phys. Rev. A]*
M. H. Cheng, K. E. Khosla, C. N. Self, M. Lin, B. X. Li, A. C. Medina, M. S. Kim

We design a Clifford-initialization algorithm that systematically improves initial-state overlap with the ground state, demonstrate scalable correlation improvements for quantum-chemistry systems at long bond lengths beyond the Hartree–Fock method, and numerically demonstrate improved trainability for the H_6 molecule.

Optimal Particle-Conserved Linear Encoding for Practical Fermionic Simulation — *[Phys. Rev. Research]*
M. H. Cheng, Yu-Cheng Chen, Qian Wang, V. Bartsch, A. C. Medina, M. S. Kim, Alice Hu, Min-Hsiu Hsieh

We identify a complexity duality between error correction and number-conserved simulation at the qubit–gate–measurement–decoding levels, enabling conserved-subspace encoding protocols on various quantum primitives. We demonstrate simultaneous CNOT cost reduction and improve convergence with subspace VQE simulation. Poster Presentation, QIP 2024 (Taipei).

Neural Network Assisted Fermionic Compression Encoding: A Lossy-QSCI Framework for Scalable Quantum Chemistry Simulations — *[Phys. Rev. Research]*
Yu-Cheng Chen, Ronin Wu, M. H. Cheng, Min-Hsiu Hsieh

We propose a quantum subspace configuration-interaction algorithm that optimizes number-conserved subspace encodings, rather than the ansatz alone. Through iterative quantum sampling and classical post-processing, our hybrid method refines ground-state estimates with high quantum-resource and computational efficiency. Selected media coverage: Quantum Zeitgeist.

Characterization of Unlearnable Noise with Generalized Cycle Benchmarking — *[In Preparation]*
M. H. Cheng, Stefano Mangini, A. C. Medina, Sergei Filippov, Matteo A. C. Rossi, M. S. Kim

We introduce the concept of quantum instruments with classical feed-forward in post-processing, leading to a new theory of Pauli noise propagation. We unify previous MCM noise-learning protocols and prove conditions for learnability of Pauli noise in Clifford gates in the presence of mid-circuit measurements. We benchmark interleaved CNOT gates and MCMs on IBM devices as a demonstration.

CITIZENSHIP

I am a Portuguese national and hold a Portuguese passport.

References available upon request